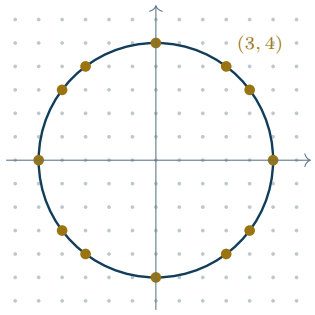
NUMBER THEORY LADDER 7 OF 9

Sums of two squares and Pythagorean triples

Counting lattice points on a circle, and squaring a Gaussian integer to build right triangles.

1 / THE RULE AND THE PICTURE

The number of integer points on the circle $x^2 + y^2 = n$ is $r_2(n) = 4(d_1 - d_3)$



$x^2 + y^2 = 25$ has 12 points

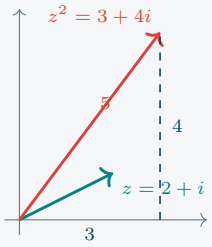
Here d_1 and d_3 count the divisors of n that are 1 and 3 modulo 4, and the factor 4 is the four units $\pm 1, \pm i$:

$$\boxed{r_2(n) = 4\big(d_1(n) - d_3(n)\big)}$$

For 25 the divisors $1, 5, 25$ are all $1 \bmod 4$, so $r_2 = 4 \cdot 3 = 12$, which are $(\pm 5, 0)$, $(0, \pm 5)$, $(\pm 3, \pm 4)$ and $(\pm 4, \pm 3)$. For 3 the divisors $1, 3$ cancel and $r_2 = 0$.

2 / EUCLID'S FORMULA / SQUARING $m + ni$

Squaring a Gaussian integer squares its length, so $(m + ni)^2$ is a right triangle



Since $(m + ni)^2 = (m^2 - n^2) + 2mn\,i$ and $|z^2| = |z|^2$,

$$\boxed{(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2}$$

With $m = 2$ and $n = 1$ the point $z^2 = 3 + 4i$ gives the $3, 4, 5$ triangle, and its angle is twice the angle of z. The triple is primitive when $\mathrm{HCF}(m, n) = 1$ and m, n have opposite parity.

3 / THE FIRST TRIPLES

Every primitive Pythagorean triple comes from exactly one pair (m, n)

(m, n)	triple (a, b, c)	check
$(2, 1)$	$(3,\ 4,\ 5)$	$3^2 + 4^2 = 5^2$
$(3, 2)$	$(5,\ 12,\ 13)$	$5^2 + 12^2 = 13^2$
$(4, 1)$	$(15,\ 8,\ 17)$	$15^2 + 8^2 = 17^2$
$(4, 3)$	$(7,\ 24,\ 25)$	$7^2 + 24^2 = 25^2$
$(5, 2)$	$(21,\ 20,\ 29)$	$21^2 + 20^2 = 29^2$
$(5, 4)$	$(9,\ 40,\ 41)$	$9^2 + 40^2 = 41^2$
$(6, 1)$	$(35,\ 12,\ 37)$	$35^2 + 12^2 = 37^2$
$(6, 5)$	$(11,\ 60,\ 61)$	$11^2 + 60^2 = 61^2$

The legs may swap, as $(4, 1)$ gives $15, 8, 17$, and for a primitive triple the hypotenuse $c = m^2 + n^2$ is odd with every prime factor $\equiv 1 \pmod 4$.

4 / COUNTING TRIPLES WITH A GIVEN HYPOTENUSE / $c = 65$

A hypotenuse needs a prime factor that is $1 \bmod 4$, and $65 = 5 \cdot 13$ has two

The triples with hypotenuse 65 are $(16, 63)$, $(33, 56)$, $(25, 60)$ and $(39, 52)$, the first two primitive and the others multiples, $5 \cdot (5, 12)$ and $13 \cdot (3, 4)$. They come from $65 = 5 \cdot 13 = (2 + i)(2 - i) \cdot (3 + 2i)(3 - 2i)$, since choosing $(2 + i)(3 + 2i) = 4 + 7i$ or $(2 + i)(3 - 2i) = 8 - i$ gives the two representations $65 = 4^2 + 7^2 = 8^2 + 1^2$, and $r_2(65) = 4 \cdot 4 = 16$ points. The primes $3 \bmod 4$ such as $3, 7, 11$ never occur in a primitive hypotenuse.

5 / WHY $(m + ni)^2$ GIVES EVERY PRIMITIVE TRIPLE

If $a^2 + b^2 = c^2$ with no common factor, then $a + bi$ is a square times a unit

Writing $c^2 = (a + bi)(a - bi)$ and noting that $a + bi$ and $a - bi$ share no prime factor in $\mathbb{Z}[i]$ when $\mathrm{HCF}(a, b) = 1$, unique factorisation forces each factor to be a unit times a square, so $a + bi = u\,(m + ni)^2$. The unit only swaps or changes the signs of the legs, which leaves the triangle unchanged.

ASIDE / RATIONAL POINTS

Dividing by c, the triple $(3, 4, 5)$ is the point $(\frac{3}{5}, \frac{4}{5})$ on the unit circle, with slope parameter $t = \frac{n}{m} = \frac{1}{2}$.

LINKS TO OTHER SHEETS

The factorisation in $\mathbb{Z}[i]$ is set up in *Primes mod 4 and the Gaussian integers*; the parametrisation by t is in *The line $ax + by = c$* and the circle sheet; the next sheet is *Primes mod 3 and the Eisenstein integers*.

Convention. $r_2(n)$ counts integer solutions of $x^2 + y^2 = n$ with signs and order; d_1, d_3 count divisors $\equiv 1, 3 \pmod 4$.

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